

**HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT**

**ATTACHMENT H**

**HOV ANCHOR DISPLACEMENT CORRESPONDENCE**

**E-mail dated October 6, 1999**

**Letter C09B2-114**

**Meeting Minutes, Friday, October 8, 1999; File No. CN-9.09B2.28.10**

**Letter C09B2-0350**

**Field Engineer's Daily Report, October 13, 1999**

**October 19 Memorandum**

**Letter C09B2-0139**

**Letter C09B2-0409**

**Field Engineer's Daily Report, November 9 and November 15, 1999**

**Letter C09B2-0436**

**Letter C09B2-0155**

**Adhesive Anchor Repair Procedure, 12/27/99**

**Letter C09B2-0499**

**Adhesive Anchor Repair Procedure, 2/16/00**

**(68 Pages)**

**E-mail dated October 6, 1999**

From: PCAIKELE  
Sent: Wednesday, October 06, 1999 7:28 PM  
To: GPBAXTER; RESTEFFY; [REDACTED]  
Cc: AVSLAGLE; RJRICHAR DJBATES  
Subject: Agenda for 10/8 technical issues mtg.

Gary, et.al.,

Listed below is the proposed agenda for the 10:00AM 10/8 technical issues mtg. divided into two parts:

Part I - resolving the differences between those items that are required by engineering & architects for the end product to be satisfactory vs. what is shown in the drawings and specifications and any discrepancies between the two.:

Issue 1 - resolve the need for locking devices between the head of the bolt and the ceiling plate where bolts are used to attach the plates to embedded struts as typically shown in detail 20 of S-276, section E of S-276, & details 1 & 3 of S-277. The SDC maintains that this issue is adequately addressed in spec 723.480-3.01-I.4. The contractor maintains that his paragraph addresses only the loosening of the "nuts" and that the ferruled nut which is installed in the embedded strut constituted a locking device for the nut. The SDC needs to confirm that the locking device is required so as to prevent the "bolt" from loosening & then we will deal with the commercial issues later.

Issue 2 - Clarify where grout or flush bearing is required between the cast in place ceiling and the hanger upper plate assembly. Note 2 of S-002 calls for . . . "non-shrink grout locally between the two surfaces in order to provide a bearing surface consistent with industry standards for concrete to steel connections." In as much as most of these connections are in tension, it would seem that the vast majority of connections would not need to be grouted as it just is not really necessary. We need to identify only those connections that would be in a compression mode and hence single them out for grouting.

Issue 3 - Note 3 on S-002 requires sealant between steel to steel surfaces at all connections and surfaces that form air plenums. We have requested the contractor to apply sealant to the angle to plate surfaces as typically shown in section A of S-275, and the contractor has balked on the basis that this note only applies to the cellular decking plenum side walls. Note that the mechanical group finds the 1/16" to 1/4" gaps at the plate to angle fit up surfaces unacceptable and wants them sealed.

Issue 4 - Is it necessary to paint charcoal gray the exposed bolts, nuts, 2"x 2" washers and exposed threads of epoxy anchors in the connection at the vertical leg of the angle at the top of the precast panels, as typically shown in section A of S-377. Similar question for the components associated with the alternative detail at transition panel of S-377, referencing the 2" dia pipe and the 3" x 3" plate washers.

Issue 5 - Address the question, "What is causing the 5/8" epoxy ceiling anchors to pullout in the ceiling module HOV079 at 4 locations? Is it a design problem or an installation problem or unknown?"

other issues as raised during the meeting

Part II

RFI's with need date in [ ]'s

#98 - core insulation for sliding doors [10/7]

#97 - ceiling hanger/dry standpipe hanger conflict [10/8]  
#92 - FCN-019, additional ceiling inserts [10/8]  
#89 - signage schedule clarification [10/22]  
RE-011 - Missing & incorrect details for egress stair 13 [10/15]  
RE-07 - installation of radio equipment boxes & associated conduits [10/15]

Submittals with need dates in [ ]'s

727.100-001B; sheet membrane waterproofing [10/8]  
728.310-001B; ceiling/wall access hatches product data [10/14]  
728.310-002A; vertical wall access panels [10/14]

FCN's for RFI's #83, #89

Items from Open action items list:

#3B; resolve construction responsibility for certain architectural features between mainline contacts and C09B2 per previously distributed list  
#78

-moving electrical pull boxes closer to roadway so that opening the lids does not interfere with the precast wall panels

Thanks,

Phil

**Letter C09B2-114**



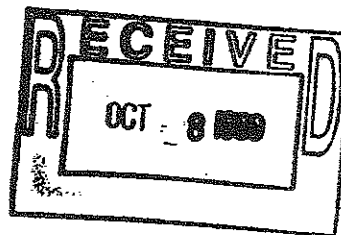
600 Memorial Drive, Cambridge, Massachusetts 02139

Telephone: (617) 864-6300 Fax: (617) 864-8766

October 7, 1999

Letter No. C09B2-114  
MCC No. 9809-0489-W

Central Artery/Tunnel Project  
Bechtel/Parsons Brinckerhoff  
185 Kneeland St.  
Boston, MA 02111



Attention: Mr. R. E. Steffy  
Authorized Representative

Subject: Central Artery (I-93) / Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
Adhesive Anchor Movement

Dear Mr. Steffy:

As noted in our field inspection of October 7, 1999, a small percentage of the adhesive anchors in the HOV ceiling mockup appear to show signs of tensile movement, thus creating gaps in the ceiling hanger plate to CIP ceiling interface. The sequence of events leading to our notification is as follows:

- 1) Ceiling module installed using proper securement prior to support removal in early August, '99. Inspection noted no movement subsequent to initial erection.
- 2) Small movement of anchor and attached plate (approx. 1/16") noted on September 9, 1999, by foreman installing plenums.
- 3) Periodic inspections by MCC revealed an increased movement to a worst case movement of 1/2" in early October of 1999, leading to notification.

Whereas these anchors have been installed and tested in accordance with contract documents, MCC respectfully requests direction pertaining to any repair or mitigative measures required.

For further correspondence on this matter please contact this office.

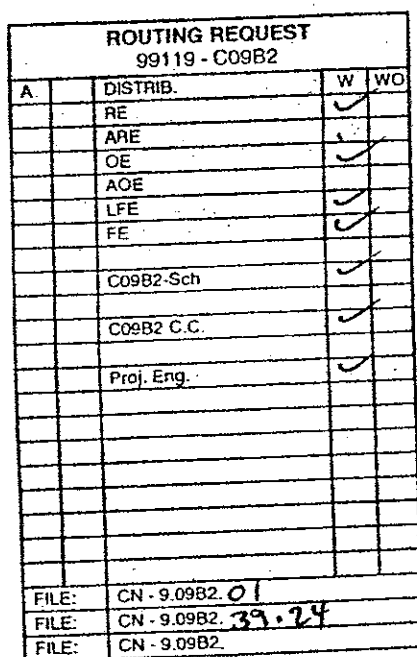
Sincerely,  
The Modern Continental Construction Company, Inc.

  
Robert Coutts  
Sr. Project Manager

cc: Martin Charney, B/PB  
C. Madden, MCC

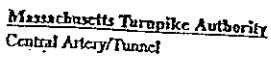
An Equal Opportunity Employer

CA/T 023981



7

**Meeting Minutes, Friday, October 8, 1999; File No. CN-9.09B2.28.10**



## 19-OCT-1999

AD010, (11/93) ORACLE (02/97)

PPM - 1007, 1303

CAT 079708



# MEETING MINUTES

Central Artery/Tunnel

|                                                    |     |                                              |
|----------------------------------------------------|-----|----------------------------------------------|
| DATE & TIME: Friday, October 8 1999                |     | File No.: CN-9.09E2.28.10                    |
| C09B2 1-90 SEAPORT ACCESS TUNNEL FINISHES          |     | LOCATION: C09B2 Field Office Conference Room |
| SUBJECT: Technical Meeting #9 - Engineering Issues |     | Page 1 of 2                                  |
| ORIGINATED BY: R. STEFFY                           |     |                                              |
| PARTICIPANTS: SEE ATTACHED LIST                    |     | RECORDED BY: P. Alkate                       |
| 0009                                               | A.1 |                                              |
| 0009                                               | A.2 |                                              |
| 0009                                               | B.1 |                                              |
| 0009                                               | B.2 |                                              |
| 0009                                               | C.1 |                                              |
| 0009                                               | C.2 |                                              |
| 0009                                               | D.1 |                                              |

Issue 1 - Resolve the need for locking devices between the head of the bolt and the ceiling plate where bolts are used to attach the plates to embedded steel as typically shown in detail 20 of S-276, section E of S-276, & detail 1 & 3 of S-277. The SDC maintains that this move is adequately addressed in spec 723.410.3.01.5.4. The contractor maintains that his paragraph addresses only the loosening of the "nut" and that the forced nut which is installed in the embedded steel constitutes a locking device for the nut. The SDC needs to confirm that the locking device is required so as to prevent the "bolt" from loosening.

SDC Discussion: Reiterated that the design requires a locking device between the bolt head and the plate where plates are attached to embedded steel in order to prevent the "locking out" of the bolt from the connection. SDC also mentioned that if the Contractor can furnish a letter from the manufacturer of the forced nut that the item will prevent the bolts from backing out, then the SDC will consider the requirement for the locking device. Contractor is required to install the locking device.

Issue 2 - Clarify where grout or flush bearing is required between the cast in place ceiling and the hanger upper plate assembly. Note 2 of S-202 calls for "... non-shrink grout locally between the two surfaces in order to provide a bearing surface consistent with highway standards for concrete to steel connections." In as much as most of these connections are in tension, it would seem that the vast majority of connections would not need to be grouted as it just is not necessary. Need to identify those connections that would be in compression mode and require grouting.

Discussion: SDC explained that grout would not be required at connections that are in tension. It would be MCC's responsibility to furnish calculations to show those connections plates that are in compression to justify requiring a flush steel to concrete connection by grouting of those plates. It was pointed out that the connections that have diagonal bracing are in both tension and compression.

Issue 3 - Note 3 on S-202 requires sealant between steel to steel surfaces at all connections and surfaces that form air plenums. The contractor is to apply sealant to the angle to plate surfaces as typically shown in section A of S-275, and the contractor has indicated that the note only applies to the cellular decking plenum side walls. Note that the mechanical group find the 1/16" to 1/4" gaps at the plate to angle fit up surfaces unacceptable.

Discussion: The steel to steel surfaces must be sealed with sealant. It was pointed out that these surfaces could be sealed from either above the ceiling plenum or below them.

Issue 4 - Is it necessary to paint charcoal gray the exposed bolts, nuts, 2"x2" washers and exposed threads of epoxy anchor in the connection at the vertical leg of the angle at the top of the precast panels, as typically shown in section A of S-277. A similar question for the components associated with the alternative detail at transition panel of S-277, referencing the 2" dia pipe and the 3" x 3" plate washers.

CLOSED

CLOSED

CLOSED



Central Artery/Tunnel

# MEETING MINUTES

|                                                    |                                    |                                              |
|----------------------------------------------------|------------------------------------|----------------------------------------------|
| DATE & TIME: Friday, October 8 1999                |                                    | File No.: CN-9.09E2.23.10                    |
| C09E2                                              | 190 SEAPORT ACCESS TUNNEL FINISHES | 10:00 AM                                     |
| SUBJECT: Technical Meeting #9 - Engineering Issues |                                    | LOCATION: C09E2 Field Office Conference Room |
| ORIGINATED BY                                      | R. STEFFY                          | Page 2 of 2                                  |
| PARTICIPANTS:                                      | SEE ATTACHED LIST                  | RECORDED BY: P. Akele                        |
| SYNOPSIS                                           | ORIGIN                             | ASSIGNED                                     |
| DATE                                               | DATE                               | DATE                                         |
| C09E2.3 ENGINEERING REQUIREMENTS                   |                                    |                                              |
| 0009                                               | D.2                                | 10/08/1999                                   |

Discussion: It was pointed out that it appeared that Note 2 for point on S-002 addressed this issue and that it would be necessary to apply the necessary coating systems to those exposed to view elements. B/FB Architectural determined after the meeting that it will be necessary to make these exposed concrete charcoal gray as required by Note 2 on S-002.

Issue 5 - "What is causing the 5/8" epoxy ceiling anchors to pullout in the ceiling module HOV079 at 4 locations?"

Discussion: It was pointed out how important it is for all of the hangers that are to support the individual modules to be attached to the module and adjusted properly so that each is taking its portion of the load before the lifting device is lowered, thereby releasing the ceiling module to be supported by its hangers. MCC explained that all of the hangers were properly attached before the lifting device was released from the module. MCC explained that all of the vendor (Tower Framework) of the epoxy used to install the subject anchor bolts in the ceiling has been notified and will be visiting the site soon to examine the problem. It is requested & expected that they will do what is necessary to determine the cause of the problem, how to remedy it and how to prevent it from recurring. SDC will re-visit the calculations to verify that the ceiling anchor system is properly designed and not overloaded. R. Steffy explained that it is MCC's responsibility to replace the failed anchors and reset them. He will be responding to MCC's letter #114 on this subject.

Issue #6 - Define "Visible" as it pertains to the plenum walls in the ceiling system and determining which should be painted/coated the charcoal gray.

Discussion: B/FB Architectural explained that the "reference point" from which to determine which side plenums are visible is for a person to stand or walk on the steps on the side of the pavement and look up & forward.

Issue #7 - MCC advised that their fireline subcontractor, Norfolk Mechanical, has turned in a visual fireline fireline alternative for those firelines that are exposed and currently require threaded fittings.

BF Discussion: B/FB & SDC responded that the alternative would be reviewed, but that chain operated isolation valves would be required every 450 ft and that it would be necessary for the subcontractor to furnish tools to prevent the opening of these valves by anyone other than the BFD.

Issue #8 - The question was raised by MCC as to what minimum end distances, if any, is required for connecting bolts to the embedded ceiling studs. It was noted that this question is similar to the issue raised in REF #102.

Discussion: SDC responded by noting that a minimum of 3 inches must be maintained between the centerline of the last bolt and the end of the embedded stud.

CLOSED

OPEN

CLOSED

OPEN

CLOSED

GANNETT FLEMING, INC.

CAIT 079710

**Letter C09B2-0350**



Massachusetts Turnpike Authority  
Central Artery/Tunnel Project

October 12, 1999

Letter No.: C09B2-0350  
File No.: CN-9.09B2.02  
CN-9.09B2.39.24

Mr. Robert Coutts  
Project Manager  
Modern Continental Companies, Inc.  
600 Memorial Drive  
Cambridge, MA 02210

Subject: Central Artery (I-93)/Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
**Adhesive Anchor Movement**

Reference: MCC Letter No. 0114, dated 10/07/99

Dear Mr. Coutts:

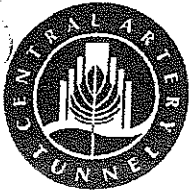
As discussed during our Engineering Issues meeting of October, 8 1999, and in response to the above referenced letter, the following was discussed.

We concur that the adhesive anchors were load tested in accordance with the design requirements, however we cannot confirm that the adhesive anchors are correctly loaded. A review of the ceiling system indicates that several turnbuckles are loose. This could induce excessive loading to the point that the anchors may be over stressed which may cause movement.

Other possibilities exist which could cause anchor bolt failure. During erection of the ceiling module the lifting system may have been released in such a manner to cause excessive stress to the anchors (too quickly) or prior to uniform loading of all turnbuckles. Failure to properly prepare & clean the drilled holes could cause a bond failure of the adhesive to the concrete over time.

The following action items were discussed and agreed to:

1. MCC to contact the supplier of the adhesive anchor bolt system, review the installation of the failed anchors and obtain a manufacturer's recommendation.
2. MCC to confirm that all ceiling hangers are correctly loaded and have uniform load.
3. MCC to investigate with the manufacture the possibility to retest the anchors to design loading after the supports are properly adjusted.



Massachusetts Turnpike Authority  
Central Artery/Tunnel Project

Mr. Robert Coutts  
B/PB Letter No.: C09B2-0350  
October 12, 1999  
Page 2 of 2

4. B/PB will confirm that ceiling load calculations are correct.

In summary, MCC shall confirm that the ceiling system is properly installed and all connection points are providing the correct support. Anchors that have moved since ceiling installation shall be abandoned and/or removed and replaced. When replacing the defective anchors, insure that proper measures are provided to support the ceiling so that other over stressed conditions are not created.

If you have any questions, please contact me at 350-7594, ext. 56.

Sincerely,

BECHTEL/PARSONS BRINCKERHOFF

[Redacted signature]

Robert E. Steffy  
Resident Engineer  
Authorized Representative of  
the Project Director

RES/ha

cc: M. Charney

**Field Engineer's Daily Report, October 13, 1999**



Central Artery/Tunnel

## FIELD ENGINEER'S DAILY REPORT

|                                                      |  |                           |                       |  |
|------------------------------------------------------|--|---------------------------|-----------------------|--|
| CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES   |  | SHIFT: 07:00-15:30        | CONTRACT DAY NO. 331  |  |
| CONTRACT NO.: 99119 - C09B2                          |  | JOB AREA: Ramp F, EB, WB  | DAY OF WEEK WEDNESDAY |  |
| CONTRACTOR: MODERN CONTINENTAL CONSTRUCTION CO. INC. |  | WEATHER: AMSNY-CLDY 65-70 | PMCLDY-RAIN 45-55     |  |

| ACTIVITY NO.<br>PAY ITEM NO. | DESCRIPTION OF WORK<br>(TYPE OF WORK, LOCATION, QUANTITIES, TESTING, ETC.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ACCEP | PHOTO |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Y/N   | Y/N   |
| 01-<br>12710/996.9<br>10     | There was one carpenter and one laborer continuing work on the ceiling module fabrication table in the roadway at EB Sta 65+00. (No references for this logistical set-up activity)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Y     | N     |
| 01-<br>13710/996.9<br>10     | One ironworker was installing ceiling hanger plates in roof Unistruts at WB Sta 68+00. (Ref Dwg S-387; Ref Spec 960.050.3.A; Ref Submtl 960.020-010-B).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Y     | N     |
| 01-<br>14640/996.9<br>10     | Three ironworkers and one operator were placing wall panels at Ramp F South wall, starting where they left off Tuesday at approx Sta 67+47 and working Eastward. By 10:00 they had placed 7 panels or 56 lf, to approx Sta 68+03, where the wall finish becomes DAT. This completes the wall panel placement on Ramp F, South side. The crew then began placing wall panels on the north side of Ramp F, starting at the West portal and working eastward. By COB they placed eleven 8 ft panels and one 6'-8" panel for 95 lf. Total production for the day (both sides) was 19 panels and 151 lf. One additional ironworker was finalizing the top connections just behind the panel placing operation. (Ref Dwg S-377, Section A; Ref Spec 723.450.2.05; Ref Submtl 723.480-008-A). | Y     | N     |
| NA/NA                        | Two MCC laborers were performing general duty in the work area, including distributing drinking water & equipment, and clean-up. (No references for this logistical support work).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Y     | N     |
| NA/NA                        | Farch met with MSP Trooper C. Midwelt to establish a traffic detail at the ERS-2 Emergency ramp for the delivery of fuel and movement of construction equipment. (Form is attached).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Y     | N     |
| NA/NA                        | DELIVERY: Four pallets (48 bags/pallet) of Dayton Superior '1107 Advantage' non-shrink grout was delivered today and stored on the north walkway of HOV @ 69+00. This material is for grouting the wall panel 'feet' through-out the job. (Ref Dwg S-377, Det 3; Ref Spec 901.610.2.01.A; Ref Submtl 901.610-001-A).                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Y     | N     |
| NA/NA                        | DELIVERY: The following ceiling hanging hardware was delivered today:<br>11 pallets- Ceiling hanger assemblies (threaded rod, clevis, turnbuckles),<br>6 Boxes 1 1/2" x 9" A325 Hanger bolts.<br>2 Boxes 1 1/2" Galv. Nuts.<br>4 Boxes 1 1/2" Galv. Jam Nuts<br>1 Box Rubber coated washers (for separation of dis-similar metals)<br>8 Buckets 1 1/2" Galv. Nuts.<br>(Ref Dwg S-277; Ref Spec 723.480.3.02.B; Ref Submtl 723.480-001-                                                                                                                                                                                                                                                                                                                                                 | Y     | N     |

CAT 010972

UNUSUAL EVENTS: ☐ ACCIDENT (A) ☐ DELAY (D) ☐ DEFICIENCY (DR) ☐ VISITOR (V)  
☐ INSTRUCTION TO CONTRACTOR (IC) ☐ OTHER \_\_\_\_\_

REPORTED BY: JAMES BARRETT

DATE: Oct. 13, 1999

REVIEWED BY: PHIL AIKELE

PAGE: 1 OF

Signature

REPORT # 3

CRE040 (11/93) ORACLE (11/97)

REPORT OF

FTM / REM - 505



|                                                      |       |                            |                       |
|------------------------------------------------------|-------|----------------------------|-----------------------|
| CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES   |       | SHIFT: 07:00-15:30         | CONTRACT DAY NO. 331  |
| CONTRACT NO.: 99119                                  | C09B2 | JOB AREA: Ramp F, EB, WB   | DAY OF WEEK WEDNESDAY |
| CONTRACTOR: MODERN CONTINENTAL CONSTRUCTION CO. INC. |       | WEATHER: AM SNY-CLDY 65-70 | PM CLDY-RAIN 45-55    |

CAIT 010971

DATE: Oct. 13, 1999

PAGE: 2 OF

REPORT # 3

REPORT OF

FIM / REM - 505

17

## **October 19 Memorandum**

To: Ray Williamson – Power's Fasteners

11 PAGES TOTAL

From: Neil Joyce – Modern Continental Construction Company

Date: October 19, 1999

Re: Epoxy Anchors

As discussed this morning, please find attached the following information regarding the adhesive anchors utilized on our project:

1. Approved Testing Procedure (Conam Inspection, 2 Pages)
2. Sketch of Mock-Up Panel Area, indicating anchor locations (CA/T, 1 Page)
3. Testing Reports for Anchors in Question (Conam Inspection, 7 Pages)

Here are a couple of thoughts:

A. The testing procedure submitted was modified slightly when implemented in the field. The applied loading was increased uniformly, rather than incrementally as explained in the submittal. The amount of load was not changed (still proofed to 3.25 Kips at each anchor). This change was discussed and approved by the Engineer verbally prior to start of field tests, and is in accordance with ASTM-E488-96.

B. General Chronology. After successful testing, the support plates were installed and fastened to the adhesive anchors. At the time of installation, the plates were observed to be flush with the ceiling. A short time after the modules were installed, a gap was observed between the concrete ceiling and support plate at five locations. The situation was monitored and the gap had increased in some areas, over a period of approximately six weeks (between August 15 and October 1). Since the time of the attached sketch (dated October 8<sup>th</sup>), there has been no additional movement in the anchors. MCC is planning to measure the anchors again prior to your arrival on Thursday morning.

C. Some general questions I have are as follows:

- Is any movement in the anchor after loading considered typical?
- Does any movement indicate an anchor failure?
- If there was a problem during installation (i.e. dust, bad epoxy mix, etc.) could the anchor pass a proof test? Would an installation problem manifest itself immediately, or could it develop over time?
- At what point in loading would the anchors become overstressed? In overstressing an anchor, is it typical to see epoxy failure or anchor elongation first?
- Loading of the anchors in the final condition is required to be at a minimum 4,000 lb. If the ceiling system loads were non-uniformly distributed among the anchors, could the anchors have become overstressed? Does the ceiling system weigh enough to cause overstressing in the anchors? Ceiling module is 40' x 12', weighs approximately 14 tons (28,000 lb), and is supported by a hanger system, attached 5' on center.

As discussed with you previously, the replacement of approximately five anchors in fifty (10%) is not of great concern to me. However, with over one thousand anchors installed, proof tested but yet to be loaded, 10% re-work is not acceptable. Please give me a call after you have reviewed this material.

I look forward to meeting Stephen on Thursday.

Conam Inspection Inc.  
2 Milbury Street  
Auburn, Massachusetts 01501  
Telephone + 508-832-5500  
Facsimile + 508-832-5557  
<http://www.conaminc.com>

July 28, 1999

Modern Continental Construction Co., Inc.

6 Necco Court

Boston, MA 02210

Attn: Neil Joyce

Re: C09B2 I-90 Seaport Access, Tension testing of anchorage  
systems embedded in the concrete tunnel structure as per  
ASTM E 488-96

**CONAM** 

Gentlemen:

Conam Inspection, Inc. which is currently performing welding inspection as per C09B1, is presenting a test procedure to perform tension testing as per C09B2 on 5/8" stainless steel anchors as installed with Pover-Rust Epoxy Injection Gel-Adhesive. The equipment used to perform the tension testing is an Enerpac ten ton 1.5' stroke centerhole jack. This calibrated jack was used in our lab to perform the same tension testing but with 1500°F on the anchor rod and concrete surface. A test fixture is being made that will allow the test to be done with an unconfined test area around the anchor. There will be three or four legs spaced at 12" apart and approximately seven inches high. The Enerpac jack will sit on top of this test stand. Rod of the same size and material as the anchor will be connected to the anchor which is embedded in the ceiling. Prior to positioning the test stand the protruding anchor rod will be marked with a permanent marker at the thread nearest to the concrete surface. Nuts will be secured to the extension rod at the top of the jack. Before tightening the two nuts be sure that the forces to be applied through the loading rod are perpendicular to the surface of the structural member section. The amount of torque or pretension applied to the anchor by the attaching nuts shall be uniform for each test.

CONAM INSPECTION INC  
ANALYST ENGINE TESTING LABORATORY DIVISION

page 2 of 2

### Tension Testing - cont.

A service load of 3.25 kips tension or 3,250 pounds is to be applied in 15% increments of the 3,250 pounds until the maximum of 3,250 pounds is reached. Each increment of 15% is equal to 488 pounds. When this increment is reached it shall be held for a two minute period before proceeding to the next increment. The following chart details the actual load as the 488 pound increments are applied:

#### Total Pounds Applied

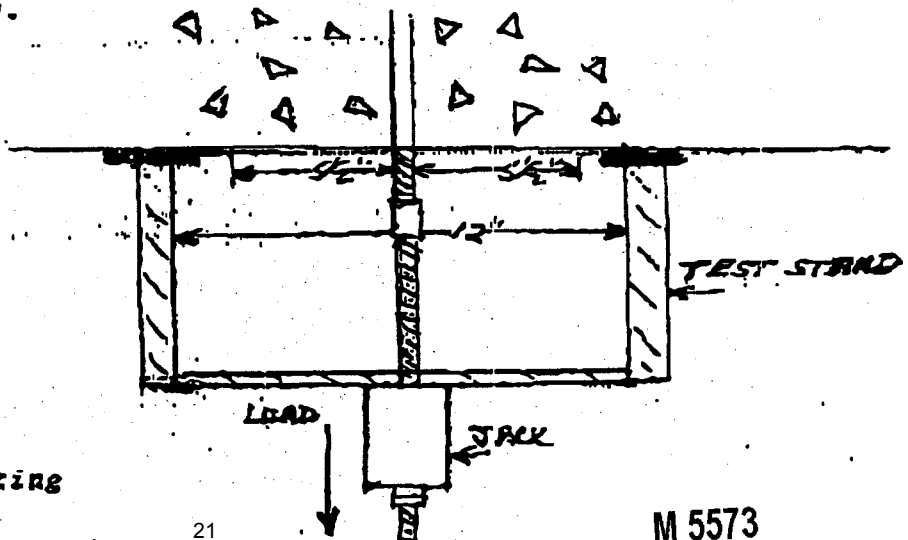
488  
976  
1464  
1952  
2440  
2928  
3250

OK  
or  
alternative  
load  
application  
(ref. ASTM  
E-488 6.2)

Remove test equipment and inspect marking on anchor. Any displacement of anchor rod as noted by the marking on the rod shall be considered a failure. A failure shall be identified by writing "FAILURE" on the ceiling adjacent to the anchor rod.

#### Unconfined Test

This illustration is how the new test stand will be built in order to provide for an unconfined test. The distance from the anchor to each leg must be a minimum of 5 1/2", therefore the total width, including the diameter of the drilled hole, is 12" between the legs.



Sincerely,  
William J. Pover  
Manager Product Testing

**Computation Sheet**

Subject HOV MOCK-UP CEILING ANCHOR  
BOLT PULL OUT MEASUREMENTS

Page \_\_\_\_\_ of \_\_\_\_\_

Made by JIM BARRETTDate 10/08/99

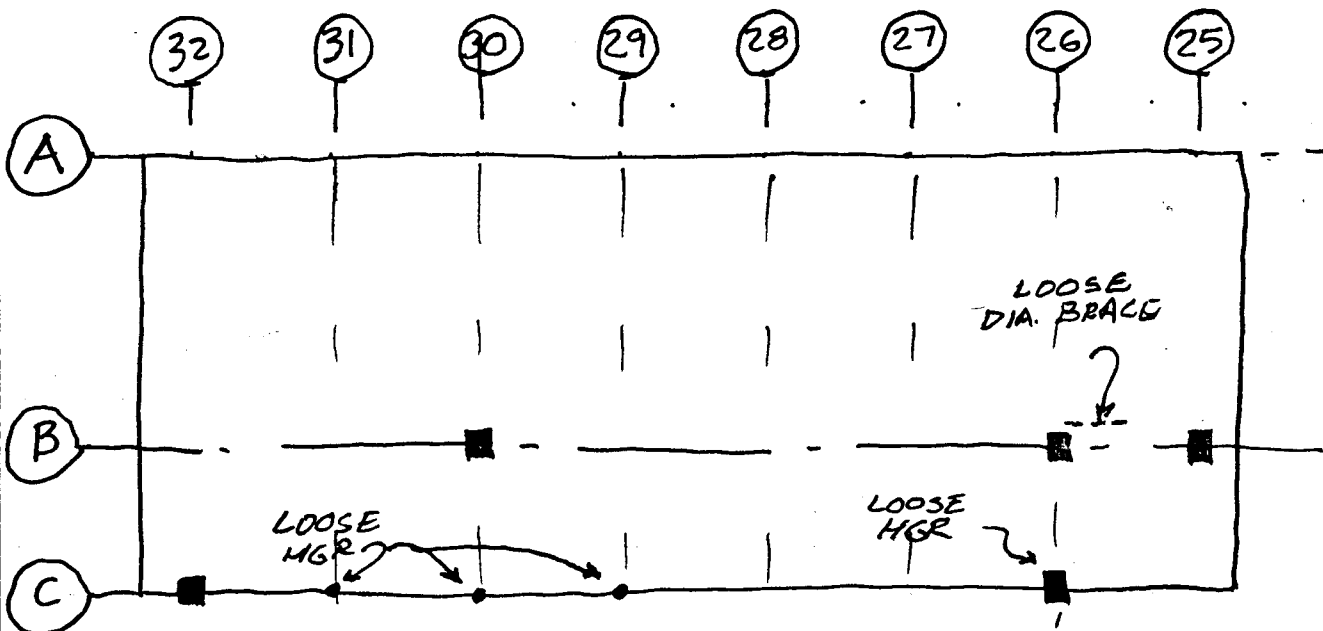
Checked by \_\_\_\_\_

Date \_\_\_\_\_

PLATE LOCATION       $\varnothing$  CORNER:      GAP BETWEEN TUNNEL ROOF AND CEILING HANGER PLATE  
NE      NW      SE      SW

|        |      |       |       |       |
|--------|------|-------|-------|-------|
| B-25   | 0    | 0     | 9/16" | 9/16" |
| B-26 * | 0    | 3/8"  | 0     | 3/8"  |
| B-30   | 1/8" | 3/16" | 0     | 1/16" |
| C-26   | 1/4" | 5/16" | 0     | 0     |
| C-32   | 0    | 0     | 3/16" | 1/8"  |

\* 4-BOLT (CROSS-BRACING)  $\varnothing$



CEILING MODULE HOV 079





**CONTINUATION SHEET  
NONDESTRUCTIVE TESTING REPORT**

CUSTOMER: MODERN CONTINENTAL  
ADDRESS: 6 NECCO CT.  
BOSTON, MA 02110  
ATTENTION: JIM MARTIN CAT #C09B2  
LOCATION: 6 NECCO CT., BOSTON, MA 02210  
JOB NO.: S4437 P.O. 9809-140642

SPECIFICATION: ASTM

STANDARD: E-488

**INSPECTION OF: ANCHOR - RESIN TO CONCRETE**

|          |        |                 |            |
|----------|--------|-----------------|------------|
| * LINE B | ROW 25 | S-1             | ACCEPTABLE |
| * LINE B | ROW 25 | N-1             | ACCEPTABLE |
| * LINE B | ROW 26 | N-1,S-2         | ACCEPTABLE |
| * LINE B | ROW 26 | N-2,S-1         | FAILURES   |
| * LINE B | ROW 30 | S-1             | ACCEPTABLE |
| * LINE B | ROW 30 | N-1             | FAILURE    |
| * LINE B | ROW 31 | S-1,S-2,N-1,N-2 | ACCEPTABLE |
| LINE B   | ROW 32 | S-1,N-1         | ACCEPTABLE |
| LINE B   | ROW 33 | S-1,N-1         | ACCEPTABLE |
| LINE B   | ROW 34 | S-1,S-2,N-1,N-2 | ACCEPTABLE |
| LINE B   | ROW 35 | S-1,N-1         | ACCEPTABLE |

INSPECTED BY:

GERARD SCHULTZ

VEL

III

DATE 7-30-99

VISUAL



M 5575



**CONTINUATION SHEET  
NONDESTRUCTIVE TESTING REPORT**

CUSTOMER: MODERN CONTINENTAL  
ADDRESS: 6 NECCO CT.  
BOSTON, MA 02110  
ATTENTION: JIM MARTIN CAT #C09B2  
LOCATION: 6 NECCO CT., BOSTON, MA 02210  
JOB NO.: S4437 P.O. 9809-140642

SPECIFICATION: ASTM

STANDARD: E-488

INSPECTION OF: ANCHOR - RESIN TO CONCRETE

|          |        |                 |            |
|----------|--------|-----------------|------------|
| LINE B   | ROW 23 | S-1,S-2         | ACCEPTABLE |
| LINE B   | ROW 24 | N-1,N-2         | ACCEPTABLE |
| LINE B   | ROW 24 | N-1,S-1         | ACCEPTABLE |
| * LINE B | ROW 26 | N-2 REPLACEMENT | ACCEPTABLE |
| * LINE B | ROW 26 | S-1 REPLACEMENT | FAILURE    |
| LINE B   | ROW 27 | S-1,N-1         | ACCEPTABLE |
| LINE B   | ROW 28 | S-1             | ACCEPTABLE |
| LINE B   | ROW 28 | N-1             | FAILURE    |

INSPECTED BY:

GERARD SCHULTZ

EL

III

DATE 8-2-99

VISUAL



M 5576



**CONTINUATION SHEET  
NONDESTRUCTIVE TESTING REPORT**

CUSTOMER: MODERN CONTINENTAL  
ADDRESS: 6 NECCO CT.  
BOSTON, MA 02110  
ATTENTION: JIM MARTIN CAT #C09B2  
LOCATION: 6 NECCO CT., BOSTON, MA 02210  
JOB NO.: S4437 P.O. 9809-140642

SPECIFICATION: ASTM

STANDARD: E-488

INSPECTION OF: ANCHOR - RESIN TO CONCRETE

|          |        |         |            |
|----------|--------|---------|------------|
| LINE C   | ROW 23 | N-1     | ACCEPTABLE |
| LINE C   | ROW 24 | N-1     | ACCEPTABLE |
| LINE C   | ROW 25 | N-1     | ACCEPTABLE |
| * LINE C | ROW 26 | N-1     | ACCEPTABLE |
| LINE C   | ROW 27 | N-1     | ACCEPTABLE |
| LINE C   | ROW 28 | N-1,N-2 | ACCEPTABLE |

INSPECTED BY:

GERARD SCHULTZ

III DATE 8-2-99  
VISUAL



M 5577



RECEIVED AUG 12 1999

## NONDESTRUCTIVE TESTING REPORT

CUSTOMER: MODERN CONTINENTAL

ADDRESS: 6 NECCO CT.

BOSTON, MA 02110

ATTENTION: JIM MARTIN

LOCATION: BOSTON, MA

JOB NO.: S4437 P.O. 9809-140642

### INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

### INSPECTION OF: ANCHOR - RESIN TO CONCRETE

|          |        |          |                          |
|----------|--------|----------|--------------------------|
| HOV      |        |          |                          |
| LINE A   | ROW 27 | N-1      | REPLACEMENT - ACCEPTABLE |
| LINE A   | ROW 30 | S-1      | REPLACEMENT - ACCEPTABLE |
| LINE A   | ROW 31 | N-1      | REPLACEMENT - ACCEPTABLE |
| * LINE B | ROW 26 | S-1      | REPLACEMENT - FAILURE    |
| LINE B   | ROW 28 | N-1      | REPLACEMENT - ACCEPTABLE |
| LINE B   | ROW 29 | N-1, S-1 | ACCEPTABLE               |
| * LINE B | ROW 30 | N-1      | REPLACEMENT - ACCEPTABLE |
| LINE C   | ROW 29 | N-1      | ACCEPTABLE               |
| LINE C   | ROW 31 | N-1      | ACCEPTABLE               |
| * LINE C | ROW 32 | N-1      | ACCEPTABLE               |

HOV 11 TOTAL  
1 FAILURE

INSPECTED BY:

GERARD SCHULTZ

LEVEL

III

DATE

8-4-99

VISUAL



RECEIVED AUG 1 2 1999

**NONDESTRUCTIVE TESTING REPORT**

CUSTOMER: MODERN CONTINENTAL  
ADDRESS: 6 NECCO CT.  
BOSTON, MA 02110  
ATTENTION: JIM MARTIN  
LOCATION: BOSTON, MA  
JOB NO.: S4437 P.O. 9809-140642

INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

INSPECTION OF: ANCHOR - RESIN TO CONCRETE - PHYSICAL TEST

|        |        |          |            |
|--------|--------|----------|------------|
| LINE C | ROW 23 | S-1      | ACCEPTABLE |
|        | ROW 24 | S-1      | ACCEPTABLE |
|        | ROW 25 | S-1      | FAILURE    |
| *      | ROW 26 | S-1      | ACCEPTABLE |
|        | ROW 27 | S-1      | ACCEPTABLE |
|        | ROW 28 | S-1, S-2 | ACCEPTABLE |
|        | ROW 29 | S-1      | ACCEPTABLE |
|        | ROW 30 | S-1      | ACCEPTABLE |
|        | ROW 31 | S-1      | ACCEPTABLE |
| *      | ROW 32 | S-1      | FAILURE    |
|        | ROW 33 | S-1      | ACCEPTABLE |
|        | ROW 34 | S-1      | ACCEPTABLE |
|        | ROW 35 | S-1      | FAILURE    |

14 TOTAL  
3 FAILURES

INSPECTED BY:

GERARD SCHULTZ 65

VEL

III

DATE 8-5-99

VISUAL

M 5579



RECEIVED AUG 25 1999

## NONDESTRUCTIVE TESTING REPORT

CUSTOMER: MODERN CONTINENTAL  
ADDRESS: 6 NECCO CT.  
BOSTON, MA 02110  
ATTENTION: JIM MARTIN  
LOCATION: BOSTON TED WILLIAMS TUNNEL  
JOB NO.: S4501 P.O. 9809 140642

### INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

### INSPECTION OF: HOV LANE

#### ANCHOR - RESIN TO CONCRETE - PHYSICAL TESTING

|          |        |     |                        |
|----------|--------|-----|------------------------|
| * LINE B | ROW 26 | S-1 | REPLACEMENT-ACCEPTABLE |
| * LINE B | ROW 30 | N-1 | SPINNER- ACCEPTABLE    |
| LINE C   | ROW 25 | S-1 | REPLACEMENT-ACCEPTABLE |
| * LINE C | ROW 32 | S-1 | REPLACEMENT-FAILURE    |
| LINE C   | ROW 35 | S-1 | REPLACEMENT-FAILURE    |

INSPECTED BY:

GERARD SCHULTZ

VEL

III

DATE 8-9-99

VISUAL



### NONDESTRUCTIVE TESTING REPORT

CUSTOMER: MODERN CONTINENTAL  
ADDRESS: 6 NECCO CT  
BOSTON, MA 02110  
ATTENTION: JIM MARTIN  
LOCATION: TED WILLIAMS TUNNEL  
JOB NO.: S4501 P.O. 9809 140642

#### INSPECTION METHOD

☐ VT ☐ MT ☐ PT ☐ UT ☐ RT ☐ ET ☒ OTHER

#### INSPECTION OF: ANCHOR RESIN TO CONCRETE - HOV LANE

|          |        |                |            |
|----------|--------|----------------|------------|
| LINE A   | ROW 16 | N1, S1         | ACCEPTABLE |
| LINE A   | ROW 17 | N1, S1         | ACCEPTABLE |
| LINE B   | ROW 15 | S1, S2, N1, N2 | ACCEPTABLE |
| LINE B   | ROW 16 | N1, S1         | ACCEPTABLE |
| LINE B   | ROW 17 | N1, S1         | ACCEPTABLE |
| LINE B   | ROW 18 | S1, S2, N1, N2 | ACCEPTABLE |
| LINE C   | ROW 15 | N1, S1         | ACCEPTABLE |
| LINE C   | ROW 17 | N1, S1         | ACCEPTABLE |
| LINE C   | ROW 18 | N1, S1         | ACCEPTABLE |
| * LINE C | ROW 32 | S1 REPLACEMENT | ACCEPTABLE |
| LINE C   | ROW 35 | S1 REPLACEMENT | ACCEPTABLE |
| LINE C   | ROW 37 | N1, S1         | ACCEPTABLE |
| LINE C   | ROW 38 | N1, S1         | ACCEPTABLE |

REVIEWED BY:

GERARD SCHULTZ  
DONALD TALLEY

LEVEL

III

DATE 8-16-99

VISUAL

INSPECTED BY:

**Letter C09B2-0139**



600 Memorial Drive, Cambridge, Massachusetts 02139

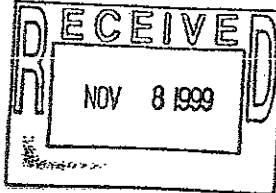
Telephone: (617) 864-6300 Fax: (617) 864-8766

MCC No. 9809-551-W

Letter No. C09B2-0139

November 8, 1999

Central Artery/Tunnel Project  
Bechtel/Parsons Brinkerhoff  
185 Kneeland Street  
Boston, MA 02111



Attention: Mr. R. E. Steffy  
Authorized Representative

Subject: Central Artery (I-93) / Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
Adhesive Anchor Movement

Reference: B/PB Letter No. C09B2-0350, dated October 12, 1999

Dear Mr. Steffy:

With regard to the adhesive anchor movement observed in Ramp HOV, Modern Continental Construction offers the following in response.

In response to the observed anchor movement, and pursuant to discussions with the Engineer regarding this matter, Modern Continental Construction invited a technical representative from Powers-Rawl Fasteners to the site in order to assess the current condition of the anchors. This site visit was conducted on October 21, 1999. A summary of this site visit and recommendations for corrective action has been attached for your review.

In the above referenced correspondence, the Engineer cites improper ceiling module installation and anchor hole preparation procedures as potential causes for the anchor movement. Modern Continental and Powers-Rawl do not agree with these statements. If the anchors were improperly installed (i.e. drill hole preparation, epoxy installation, etc.), it is likely that the anchors would have failed the proof test. As all of the anchors passed proof testing prior to module installation, it is not likely that anchor installation procedures were the cause of the anchor movement. Further, the ceiling modules were erected in accordance with the approved erection procedure, and in the presence of the Engineer. The assertion that the lifting devices were removed too quickly is unfounded.

Modern Continental Construction is currently conducting a review of the engineering associated with the adhesive anchors. Given the parameters associated with the ceiling design (approximate module weight, number of anchors supporting the module, and tensile capacity of the anchor) and the applied factors of safety, it is improbable that the anchors were overstressed as a result of erection loads or the erection procedures. While the measured torque on some anchors exceeded recommended values, the manufacturer states that these anchors are functional, and should be subjected to a proof-load test for verification.

*An Equal Opportunity Employer*

CA/T 082296

Letter No. C09B2-0139

Page 2 of 2

In summary, Modern Continental Construction confirms that all elements of the ceiling have been installed in accordance with the Contract Documents. Per the recommendations of Powers-Rawl, Modern Continental Construction will remove and replace the anchors which exhibited movement, and will proof test the anchors subjected to torque values higher than the manufacturer's recommended values in order to verify the anchor capacity. Modern Continental is currently evaluating available personnel and resources necessary to undertake this effort, and will inform the Project prior to the start of this operation.

However, in that the anchors have already passed the proof test and the root cause of the anchor failure has yet to be determined, Modern Continental considers this additional testing and support of this operation to be a change to the Contract. In accordance with Division I, Section 4.03, Modern Continental Construction reserves our rights to claim for additional costs associated with this change.

Sincerely,

Modern Continental Construction Company Inc.

  
Robert Coutts  
Sr. Project Manager

RC/NJ

cc: M. Charney, B/PB  
C. Madden  
J. Pastore

CA/T 082297

# **Powers** FASTENERS

Friday, October 28, 1999

The Modern Continental Construction Co., Inc.  
6 Necco Court  
Boston MA 02210

Attn: Mr. Neil Joyce  


Powers Fasteners, Inc.  
2 Powers Square  
New Rochelle, NY  
USA 10801

Telephone: 1-(914) 235-8300  
Fax: 1-(914) 876-6483

Web Site: [www.powers.com](http://www.powers.com)  
E-Mail: [williamson@powers.com](mailto:williamson@powers.com)

Re: CO982 I-90 Support Access

Dear Mr. Joyce:

Thank you for allowing Powers' Regional Representative, Mr. Jack Armour and Powers' Field Engineering Manager, Mr. Stephen Rice access to the above mentioned job site to inspect the six anchors that showed movement in the HOV ceiling mock up.

The following job site information was provided to Powers Fasteners by others:

- a) over (1000) 5/8" diameter Power-Fast adhesive anchors were installed in diamond cored anchor holes to a minimum embedment depth of 5"
- b) the anchor rod material was Type 316 Stainless Steel threaded rod
- c) anchor seal plugs were implemented as part of the adhesive anchoring system per installation recommendations due to overhead anchorage locations
- d) base material was reported to be solid, normal weight concrete having a minimum compressive strength of 4,000 psi
- e) adhesive anchors were installed in July of 1999.
- f) within two or three days after installation, all of the adhesive anchors were proof load tested to a load of 3,250 lbs in tension in accordance with procedures approved by the Project Engineer that followed the guidelines of ASTM E-488.

Conam Inspection, Inc.'s Field Test Reports dated July 30 through August 16, 1999 had been reviewed prior to Mr. Rice's arrival to the site. Mr. Rice was told while on site that anchors reported as failures in Conam Inspection's Reports were replaced, retested and approved prior to installation of the HOV ceiling mock up.

In Mr. Robert Courtis of Modern Continental's letter of October 7, 1999 addressed to Mr. R.E. Steffy representing the Central Artery Tunnel Project, it was reported that approximately one month after the HOV ceiling mock up installation, initial movement of about 1/16" was observed on the six anchors in question. This condition was monitored for the next six weeks whereas the worst case of anchor movement was reported to be approximately 9/16". This information also corresponds to the condition reported in your

CA/T 082298

letter of October 19, 1998. No additional anchor movement since October 19<sup>th</sup> has been reported.

Upon visual inspection of the anchors in question, Mr. Rice inquired as to details of the anchor installation as well as the installation procedure of the ceiling hanger plates. The main question raised, "Why would an anchor that was successfully proof loaded and believed to be correctly installed, move after the ceiling hanger plates were installed?"

Mr. Rice reported that he proceeded to check the applied torque on the nuts of the anchors with a torque wrench. Mr. Rice checked both the anchors that showed movement as well as anchors that had not moved and got torque readings of 120 foot-pounds. This torque value is greater than the published maximum torque range of a 5/8" diameter Power-Fast adhesive anchor (50-90 foot-pounds). The maximum torque ranges published for Powers' injection adhesive systems are based on testing performed in normal weight concrete at nine diameters of embedment depth.

Answers to your general questions of October 19<sup>th</sup> are as follows:

- Please refer to the attached excerpt from Powers' *Fastening Systems Design Manual, Second Edition*, Section 22.4.1 entitled "Displacement". Curve # 1 in the illustration shows the typical performance of an adhesive type anchor. If an appropriate factor of safety is applied to the ultimate load capacity of the adhesive, then displacement should not occur provided that proper installation procedures were employed.
- Any movement in the anchor may be an indication that the allowable load capacity of that particular installation has potentially been exceeded. As shown in Curve # 1, when displacement becomes more significant, the anchor approaches its ultimate failure load capacity.
- If a problem during installation occurred, it is possible that an anchor could pass a proof load test provided a large enough safety factor were employed. Powers Fasteners recommends a minimum safety factor of four but it appears that a higher safety factor has been used based on comparing the proof load of your field tests to the published ultimate tensile bond strength. This conservative design technique is not unusual. Adhesive anchoring problems typically reveal themselves immediately in a testing environment and do not manifest themselves over periods of time. For example, if the problem were dust in the hole, then the anchor would probably have moved during the proof loading procedure. If not properly mixed, the adhesive typically would not fully cure and remain somewhat tacky. This would also be apparent during the proof loading procedure.
- Overstress is a term more often associated with the applied torque to an adhesive anchor. The condition where the nuts have possibly been overtorqued is described as an overstressed condition. In an ultimate adhesive anchor failure condition, the mode of failure can either be base material failure or more commonly, bond failure at the interface of the adhesive and the walls of the anchor hole. Anchor rod elongation is more likely to occur in deeper embedment depths (greater than 5") where the strength of the anchor rod material has effectively been developed.
- Since Powers Fasteners, Inc. is not the designer of this ceiling hanger plate assembly, an answer to this final question cannot be provided. Powers is not

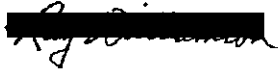
CA/T 082299

aware of the different loads that may be acting on these anchors. In addition to dead loads, live loads such as wind, equipment and, personnel that potentially affect the applied load capacity should be considered.

While we believe that the existing anchors subjected to torques exceeding our recommended range may be acceptable, the only way to verify their suitability is by performing a proof load test. Successful proof load test results would indicate that the anchors are still functional. Nuts should then be reapplied to tested anchors with a torque that is within our recommended torque range.

Should you have any questions or require further information, please feel free to contact me at (914) 235-6300. Thank you for your continued interest in Powers Fasteners.

Very truly yours,  
Powers Fasteners, Inc.



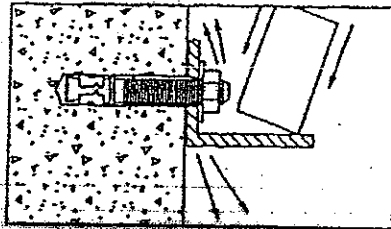
Ray Williamson  
Engineering Manager

cc: William H. Alderman, P.E. - Powers Fasteners, Inc.  
Stephen Rice - Powers Fasteners, Inc.  
Jack Armour - Powers Fasteners, New England  
Rich Tamulionis - Newman Renner Colony

CA/T 082300

## Anchor Selection Guidelines

Shock loads are instantaneous, periodic loads of high intensity such as those applied by an automobile striking a guard rail support or a truck hitting a dock bumper.



Shock load

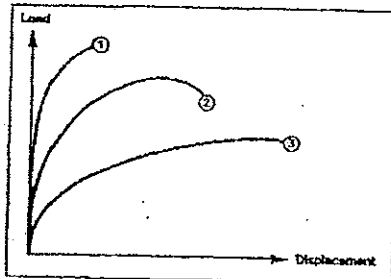
Standard industry practice with regard to safety factors varies depending upon the frequency and intensity of the load. However, safety factors for dynamic or shock load conditions may be as high as 10:1.

### 22.4 Anchor Behavior

The selection and specification of an anchor requires an understanding of basic anchor behavior or performance. A variety of performance attributes can be expected depending upon the type or style of anchor.

#### 22.4.1 Displacement

As an anchor is loaded to its ultimate (failure) load capacity, displacement or movement of the anchor relative to the base material will occur. The amount of displacement will be effected by the anchor preload, the anchor material strength, the design of the expansion mechanism, and the strength of the base material. Typical load versus displacement curves are shown in the following diagram for three anchor types.



Curve 1 shows the typical performance of an adhesive type anchor. These anchors normally exhibit elastic behavior up to the ultimate load capacity. Performance will vary depending upon the type of adhesive used, the base material strength, and the strength of the anchor rod. A deformation controlled anchor such as a steel drop-in anchor may also exhibit this type of behavior although the ultimate load capacity will normally be

much less than that of an adhesive anchor. The compression force developed by a steel drop-in is usually very high when compared to a torque controlled anchor resulting in low displacement characteristics.

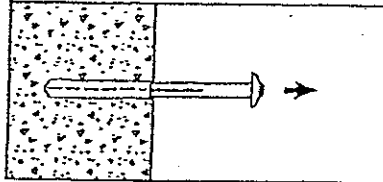
Typical performance of a torque controlled anchor such as the Power-Bolt or Power-Stud is shown in Curve 2. Displacement begins to occur after the initial preload in the anchor has been exceeded until the ultimate load capacity is achieved.

Anchors for use in light duty applications often exhibit the behavior shown in Curve 3. Once the working load has been exceeded, the anchor begins to displace or stretch until failure occurs.

#### 22.4.2 Modes Of Failure

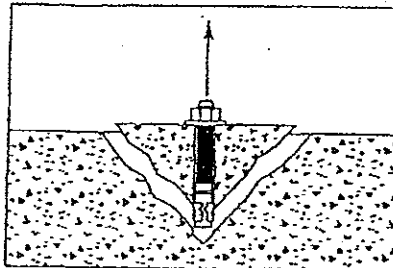
As an anchor is loaded to its ultimate capacity, the following modes of failure can occur.

##### 22.4.2.1 Anchor Pullout



This type of failure occurs when the applied load is greater than the friction or compressive force developed between the anchor body and the base material. The anchor is unable to fully transfer the load to develop the strength of the base material. For adhesive anchors, this can occur with products which have a low bond strength.

##### 22.4.2.2 Base Material Failure



When the applied load is greater than the strength of the base material, the material pulls out or fails. In concrete, a shear cone will be pulled, usually for anchors installed at a shallow embedment in the range of 4 to 5 anchor diameters of depth. The angle of the shear cone has been assumed to be 45°, however, this can vary up to 60° depending upon the anchor style and embed-

**Letter C09B2-0409**



Massachusetts Turnpike Authority  
Central Artery/Tunnel Project

November 11, 1999

Letter No.: C09B2-0409  
File No.: CN-9.09B2.02  
CN-9.09B2.39.24

Mr. Robert Couffts  
Project Manager  
Modern Continental Companies, Inc.  
600 Memorial Drive  
Cambridge, MA 02210

Subject: Central Artery (I-93)/Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
Adhesive Anchor Movement

Reference: MCC Letter No. 0114, dated 10/07/99  
B/PB Letter No. 0350, dated 10/12/99  
MCC Letter No. 0139, dated 11/08/99

Dear Mr. Couffts:

Receipt of the above referenced letter in which Modern Continental Construction considers additional testing of adhesive anchors as a change to the Contract, is hereby acknowledged.

We do not concur with your position that all elements of the ceiling have been installed in accordance with the Contract. Whereas, adhesive anchors are over torqued and the ceiling hangers are not loaded uniformly. Until the anchors in question are removed and inspected can a determination be made of correct or incorrect installation techniques.

We concur with the recommendations in your letter from Power Fasteners that Modern Continental Construction will remove and replace anchors which exhibited movement and the proof testing of the anchors subjected to torque values higher than the manufacturer's recommended values.

To summarize, the action items from our October 12, 1999 letter:

- 1 MCC to contact the supplier of the adhesive anchor bolt system, review the installation of the failed anchors and obtain a manufacturer's recommendation.

Action has been completed, Power Fasteners has reviewed the installation and provided a recommended corrective action.

185 Kneeland Street • Boston • Massachusetts • 02111 • Phone 617-951-6000 • Fax 617-457-8198

CA/T 079906

Contract No. 99119-C09B2

Letter No. C09B2-0409

Page 2 of 2

2. MCC to confirm that all ceiling hangers are correctly loaded and have uniform load.

Open Item

3. MCC to investigate with the manufacture the possibility to retest the anchors to design loading after the supports are properly adjusted.

Action has been completed. Power Fasteners has reviewed the installation and provided a recommended corrective action to retest anchors that have not indicated movement.

4. B/PB will confirm that ceiling load calculations are correct.

Action has been completed. B/PB has re-confirmed that design loads for the ceiling system are correctly stated.

Modern Continental Construction is requested to provide a schedule for the removal and replacement of the damaged anchors. B/PB requests to be present for the removal and retesting of the anchors in question. Prior to commencing new ceiling installations where adhesive anchors are used, confirm that no over torquing has occurred with previously installed ceiling plates.

Based on the above, the project has determined that there is no merit to MCC's claim for additional compensation and hereby inform you that this claim is denied.

If you have any questions, please contact me at [REDACTED]

Sincerely,

BECHTEL/PARSONS BRINCKERHOFF

[REDACTED]  
Robert E. Steffy  
Resident Engineer  
Authorized Representative of  
the Project Director

RES/ks

cc: M Charney

CA/T 079907

**Field Engineer's Daily Report, November 9 and November 15, 1999**



Central Artery/Tunnel

## FIELD ENGINEER'S DAILY REPORT

CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES SHIFT: 07:00-15:30 CONTRACT DAY NO. 358  
 CONTRACT NO.: 99119 -- C09B2 JOB AREA: EB WB HOV DAY OF WEEK TUESDAY  
 CONTRACTOR: MODERN CONTINENTAL CONSTRUCTION CO. INC. WEATHER: AMCLDY 58-63 PMCLDY 45-52

| ACTIVITY NO./<br>PAY ITEM NO. | DESCRIPTION OF WORK<br>(TYPE OF WORK, LOCATION, QUANTITIES, TESTING, ETC.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ACCEP | PHOTO |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Y/N   | Y/N   |
| 01-<br>11950/998.1<br>02      | One electrician was installing conduit in the CMU walls being built on the HOV side of CP-26. (Ref Dwg E-301; Ref Spec 998.050.2.07; Ref Submtl 998.050-011-A & 998.050-016-A)                                                                                                                                                                                                                                                                                                                                                                                                         | Y     | N     |
| 01-<br>13710/996.9<br>10      | Four ironworkers were installing ceiling hanger plates and hangers in the WB bore at approx Sta 70+00. (Ref Dwg S-387; Ref Spec 960.050.3.A; Ref Submtl 960.020-010-B)                                                                                                                                                                                                                                                                                                                                                                                                                 | Y     | N     |
| 01-<br>13720/996.9<br>10      | Six ironworkers, one teamster and one operator were hanging ceiling modules in WB. By COB they had hung 5 modules (WBD075B, WBD076A, WBD076B, WBD077A & WBD077B) for a total of 2400 sqft. (Ref Dwg S-269; Ref Specs 723.480 & 960.020; Ref Submtls 723.480-004-A & 960.020-010-B)                                                                                                                                                                                                                                                                                                     | Y     | N     |
| 01-<br>13950/998.1<br>02      | Two MEC Electricians were installing interim lighting under the ceiling modules in WB. (Ref Spec 721.520.2.01.A9; Ref Submtl 721.520-001-A)                                                                                                                                                                                                                                                                                                                                                                                                                                            | Y     | N     |
| NA/996.910                    | DELIVERY: Ten precast concrete ceiling panels were delivered today and stored near the EB fabrication area. (Ref Dwg S-345; Ref Spec 723.480; Ref Submtl 723.)                                                                                                                                                                                                                                                                                                                                                                                                                         | Y     | N     |
| NA/999.065                    | MSP Trooper W. DiNatale established a police detail at the ERS2 emergency ramp for delivery of fuel and movement of construction equipment. Form is attached.                                                                                                                                                                                                                                                                                                                                                                                                                          | Y     | N     |
| NA/NA                         | Two MCC laborers were providing general logistical support in the construction area, including clean-up and distribution of drinking water, tools and equipment. (No references apply).                                                                                                                                                                                                                                                                                                                                                                                                | Y     | N     |
| NA/NA                         | Farch met with J. Martin of MCC (QC) to verify the torque values of epoxied ceiling anchor bolts in the WB 4A2 area. Farch randomly selected 66 bolts (one third of the installed bolts) for testing. The random sample included approximately 22 from each of the three longitudinal lines of installed bolts. All tested bolts were found to require approx 1/8th turn to 'click' the torque wrench, which was set at 75 ft-lbs. The manufacturer's recommended torque for 5/8" bolts is 50 to 90 ft-lbs, hence 75 ft-lb is acceptable. (Ref Dwg. S-387; No Specs or Submtls apply). | Y     | N     |
| SUM170/700.<br>531            | One laborer was installing a fireproof polyethylene curtain at the WB portal for retention of temporary heating. (No references apply).                                                                                                                                                                                                                                                                                                                                                                                                                                                | Y     | N     |

CAIT 040506

## UNUSUAL EVENTS:

☐ ACCIDENT (A)☐ DELAY (D)☐ DEFICIENCY (DR)☐ VISITOR (V)☐ INSTRUCTION TO CONTRACTOR (IC)☐ OTHER

REPORTED BY: JAMES BARRETT

REVIEWED BY: PHIL AIKELE

Signature

DATE: Nov. 09, 1999

PAGE: 1 OF

REPORT # 3

REPORT 1 OF 3

FIN / REM - 505

CRE040 (11/93) ORACLE (11/97)



Central Artery/Tunnel

## FIELD ENGINEER'S DAILY REPORT

CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES SHIFT: 07:00-15:30 CONTRACT DAY NO. 364  
 C RACT NO.: 99119 -- C09B2 JOB AREA: EB, WB, Ramp F DAY OF WEEK MONDAY  
 CONTRACTOR: MODERN CONTINENTAL CONSTRUCTION CO. INC. WEATHER: AM 39-45 WD 15-22 PM 25-33 WD 15-20

| ACTIVITY NO./<br>PAY ITEM NO. | DESCRIPTION OF WORK<br>(TYPE OF WORK, LOCATION, QUANTITIES, TESTING, ETC.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ACCEP | PHOTO |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Y/N   | Y/N   |
| 01-<br>12710/996.9<br>10      | One ironworker was installing ceiling hanger plates and locking devices in the EB bore near Ramp A gore. (Ref Dwg S-387; Ref Spec 960.050.3.A; Ref Submtl 960.020-010-B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Y     | N     |
| 01-13720-<br>996.910          | Six ironworkers, one operator and one teamster were hanging ceiling modules in WB. By COB they had hung modules WBD082A, WBD083B, WBD083A and WBD084B, a total of 4 modules for 1920 sq ft. (Ref Dwg S-269; Ref Specs 723.480 & 960.020; Ref Submtls 723.480-004-A & 960.020-010-B)                                                                                                                                                                                                                                                                                                                                                                                                                                         | Y     | N     |
| 01-<br>13720/996.9<br>10      | Four ironworkers and one operator were assembling ceiling modules in EB for installation in WB. (Ref Dwg S-269; Ref Specs 723.480 & 960.020; Ref Submtls 723.480-004-A & 960.020-010-B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Y     | N     |
| 01-<br>13720/996.9<br>10      | Two ironworker were installing ceiling hangers in EB 4A2 area. (Ref Dwg S-387; Ref Spec 960.050; Ref Submtl 960.020-010-B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Y     | N     |
| 01-<br>14640/996.9<br>10      | One laborer was grouting the feet of wall panels in Ramp F, North wall. By 11:00 he had grouted all but the last panel, which was intentionally left ungrouted in case adjustment is needed when wall panel placement continues.<br>The grouting operation moved to the North wall of WB bore, just east of CP-29 at the false wall. By COB, he had completed the false wall area. (Ref Dwg S-377; Ref Spec 901.610; Ref Submtl 901.610-001-A)                                                                                                                                                                                                                                                                              | Y     | N     |
| FAB760/996.<br>910            | Three Stainless steel sliding door frames were delivered to the WB bore today. The unassembled frames were placed flat on the walkway and filled with non-shrink grout. The frames did not have an adhesive backed protective film as required by Spec 728.314.1.05.A. N. Joyce of MCC stated that this was because they were prototypes, and the production doors will be so protected. Joyce also asked FArch if this method of grouting was acceptable. FArch stated that the technique should be the same as is used in the fire rating test. Joyce said the at least one of these frames will be sent to the laboratory for 2-hour fire rating tests. (Ref Dwg H-904; Ref Spec 728.314; Ref Submtl 728.314-001-B & C). | Y     | N     |
| NA/NA                         | DELIVERY: There were 10 ceiling panels delivered to the site today and stored in the EB Fabrication area. (Ref Dwg S-345; Ref Spec 723.480; Ref Submtl 723.480-008-A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Y     | N     |
| VA/NA                         | Two laborers were performing general logistical duties including clean-up and distribution of water, tools and equipment. (No references apply.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Y     | N     |

## UNUSUAL EVENTS:

☐ ACCIDENT (A) ☐ DELAY (D) ☐ DEFICIENCY (DR)  
☐ INSTRUCTION TO CONTRACTOR (IC) ☐ OTHER \_\_\_\_\_

CA/T 011078

☐ VISITOR (V)

REPORTED BY: JAMES BARRETT

DATE: Nov. 15, 1999

VIEWED BY: PHIL AIKELE

Signature

PAGE: 1 OF

REPORT # 3

REPORT OF

RE040 (11/93) ORACLE (11/97)

FIM / REM - 505



Central Artery/Tunnel

# FIELD ENGINEER'S DAILY REPORT

| TRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SHIFT: 07:00-15:30        |              | CONTRACT DAY NO. 364 |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------|----------------------|--|
| CONTRACT NO.: 99119 C09B2                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | JOB AREA: EB, WB, Ramp F  |              | DAY OF WEEK MONDAY   |  |
| CONTRACTOR: MODERN CONTINENTAL CONSTRUCTION CO. INC.                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | WEATHER: AM 39-45 WD15-22 |              | PM 25-33 WD15-20     |  |
| ACTIVITY NO./<br>PAY ITEM NO.                                                                                                                                                                                                                          | DESCRIPTION OF WORK<br>(TYPE OF WORK, LOCATION, QUANTITIES, TESTING, ETC.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ACCEP<br>Y/N              | PHOTO<br>Y/N |                      |  |
| NA/NA                                                                                                                                                                                                                                                  | FArch met with J. Martin of MCC (QC) to verify the torque values of epoxied ceiling anchor bolts in the EB 4A2 area. FArch randomly selected 64 bolts (25% of installed bolts) for testing. The random sample included approximately 16 bolts from each of the four longitudinal lines of installed bolts. All tested bolts were found to require between 1/8 and 1/4 turn to 'click' the torque wrench, which was set at 75 ft-lbs. The manufacturer's recommended torque for 5/8" bolts is 50 to 90 ft-lbs, hence 75 ft-lbs is acceptable. (Ref Dwg S-387, Ref Spec 723.480.3.07.A; Ref Submtl 723.480-014-B) | Y                         | N            |                      |  |
| NA/NA                                                                                                                                                                                                                                                  | DELIVERY: Two truck loads of ceiling structural framing steel were delivered to the boatsection over the weekend (Ref Dwg S-269; Ref Spec 723.480.3.02.B; Ref Submtl 723.480-001-A)                                                                                                                                                                                                                                                                                                                                                                                                                             | Y                         | N            |                      |  |
|                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |              |                      |  |
| CAT 011079                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |              |                      |  |
| UNUSUAL EVENTS:                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |              |                      |  |
| <input type="checkbox"/> ACCIDENT (A) <input type="checkbox"/> DELAY (D) <input type="checkbox"/> DEFICIENCY (DR) <input type="checkbox"/> VISITOR (V)<br><input type="checkbox"/> INSTRUCTION TO CONTRACTOR (IC) <input type="checkbox"/> OTHER _____ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |              |                      |  |

REPORTED BY: JAMES BARRETT

DATE: Nov. 15, 1999

VIEWED BY: PHIL AIKELE

Signature

PAGE: 2 OF

REPORT # 3

REPORT OF

RE040 (11/93) . ORACLE (11/97)

43

FIM / REM - 505

**Letter C09B2-0436**



Massachusetts Turnpike Authority  
Central Artery/Tunnel Project

December 2, 1999

Letter No.: C09B2-0436  
File No.: CN-9.09B2.02  
CN-9.09B2.49.01.001

Mr. Robert Coutts  
Project Manager  
Modern Continental Companies, Inc.  
600 Memorial Drive  
Cambridge, MA 02210

Subject: Central Artery (I-93)/Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
Deficiency Report (DR) No. 001  
Adhesive Anchor Failures

Dear Mr. Coutts:

Attached for your action is Deficiency Report Number 001 (DR-001) which addresses the failure of adhesive anchors in the HOV ceiling. Please review DR-001 and provide a response to this office by the close of business on December 9, 1999.

If you have any questions or concerns, please contact me at [REDACTED]

Sincerely,

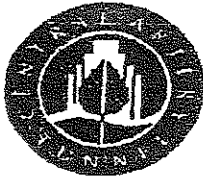
BECHTEL/PARSONS BRINCKERHOFF

[REDACTED]  
Robert E. Steffy  
Resident Engineer  
Authorized Representative of  
the Project Director

RES/DEH/dh

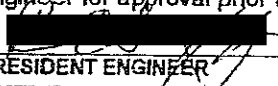
Attachment: DR-001 w/Sketch (2 pages)

cc: M. Charney  
G. Baxter



## DEFICIENCY REPORT

CONTRACT NO: 99119-C09B2 CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES  
DATE OF DEFICIENCY: December 1, 1999 DEFICIENCY REPORT NO: 001  
DATE ISSUED: December 1, 1999 FIELD ENGINEER: P. C. Aikele

|                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| TO:<br>Bob Coutts                                                                                                                                                                                                                                                                                                                            | DISCUSSED WITH:<br>Bill Britto & Jim Martin                                                                                      |
| SPECIFICATION SECTIONS:<br>723.480 - 2.08                                                                                                                                                                                                                                                                                                    | SUBMITTAL NO:<br>N/A                                                                                                             |
| SCHED. ACTIVITY NO:<br>N/A                                                                                                                                                                                                                                                                                                                   | PAY ITEM NO:<br>N/A                                                                                                              |
| REQUIREMENT:<br>The ceiling adhesive anchors shall be installed according to the project specifications 723.480-2.08 and manufacturer's requirements so that when the ceiling system is attached to them they do not fail, i.e. pull out.                                                                                                    |                                                                                                                                  |
| DEFICIENCY:<br>At least five ceiling adhesive anchors in the area of ceiling module HOV 079 have failed, that is pull-out has been observed, since the hanging of the ceiling module. See the attached sketch for the locations of the failed anchors and the amount of pull-out that has been observed.                                     |                                                                                                                                  |
| RECOMMENDED ACTION:<br><br>(1) MCC shall contact the adhesive anchor manufacturer to arrange for a site visit and investigation to determine the reason(s) for the failure of the anchors and the proposed corrective actions.<br>(2) MCC shall submit the proposed corrective actions to the Engineer for approval prior to implementation. | RESPONSE DUE DATE: <u>12/08/99</u>                                                                                               |
|                                                                                                                                                                                                                                                                                                                                              | <br>RESIDENT ENGINEER<br>DATE <u>12-2-99</u> |
| CORRECTIVE ACTION TAKEN:                                                                                                                                                                                                                                                                                                                     | FORECAST COMPLETION DATE:                                                                                                        |
| CONTRACTOR QUALITY CONTROL MANAGER OR APPROPRIATE CONTRACTOR REPRESENTATIVE _____ DATE _____                                                                                                                                                                                                                                                 |                                                                                                                                  |
| DESIGN CHANGE: _____ YES _____ NO                                                                                                                                                                                                                                                                                                            | PROJECT ENGINEER _____ DATE _____                                                                                                |
| CORRECTIVE ACTION VERIFIED:                                                                                                                                                                                                                                                                                                                  | ACTUAL COMPLETION DATE:                                                                                                          |
| RESIDENT ENGINEER _____ DATE _____                                                                                                                                                                                                                                                                                                           |                                                                                                                                  |



CENTRAL ARTERY/TUNNEL

## Computation Sheet

Subject HOV MOCK-UP CEILING ANCHOR  
BOLT PULL OUT MEASUREMENTS

Page \_\_\_\_\_ of \_\_\_\_\_

Made by JIM BARRETTDate 10/08/99

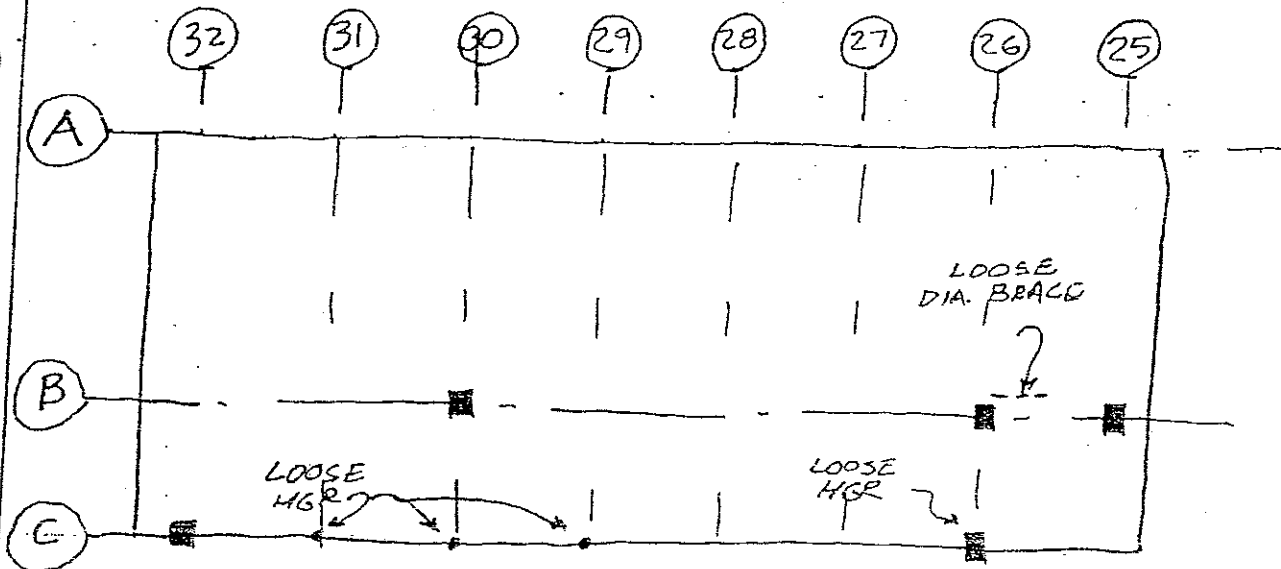
Checked by \_\_\_\_\_

Date \_\_\_\_\_

| PLATE<br>LOCATION | FE<br>CORNER: | GAP BETWEEN TUNNEL ROOF<br>AND CEILING HANGER PLATE |    |    |    |
|-------------------|---------------|-----------------------------------------------------|----|----|----|
|                   |               | NE                                                  | NW | SE | SW |

|        |  |      |       |       |       |
|--------|--|------|-------|-------|-------|
| B-25   |  | 0    | 0     | 9/16" | 9/16" |
| B-26 * |  | 0    | 3/8"  | 0     | 3/8"  |
| B-30   |  | 1/8" | 3/16" | 0     | 1/16" |
| C-26   |  | 1/4" | 5/16" | 0     | 0     |
| C-32   |  | 0    | 0     | 3/16" | 1/8"  |

\* 4-BOLT (CROSS-BRACING) FE



CEILING MODULE HOV 079



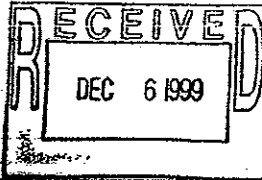
**Letter C09B2-0155**



■ Telephone: (617) 864-6300 Fax: (617) 864-8766

Letter No. C09B2-0155

**Central Artery/Tunnel Project  
Bechtel/Parsons Brinkerhoff  
185 Kneeland Street  
Boston, MA 02111**



Subject: Central Artery (I-93) / Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
Deficiency Report (DR) No. 001 – Adhesive Anchor Failures

Dear Mr. Steffy:

In response to Deficiency Report No. 001 regarding Adhesive Anchor Failures in the area ceiling module HOV 079, attached you will find, based on the recommended actions, the corrective actions taken by MCC to resolve this issue.

If you have any questions, please do not hesitate to contact our office.

Sincerely,  
Modern Continental Construction Company Inc.

Robert Coultis  
Sr. Project Manager

RC/JM

cc: M. Charney, B/PB  
C. Madden  
J. Pastore

[illegible]

### *An Equal Opportunity Employer*

CA/T 082456



## DEFICIENCY REPORT

CONTRACT NO: 99119-C09B2 CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES  
DATE OF DEFICIENCY: December 1, 1999 DEFICIENCY REPORT NO: 001  
DATE ISSUED: December 1, 1999 FIELD ENGINEER: P. C. Aikele

|                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| TO:<br>Bob Coutts                                                                                                                                                                                                                                                                                                                            | DISCUSSED WITH:<br>Bill Britto & Jim Martin                                                                                             |
| SPECIFICATION SECTIONS:<br>723.480-2.08                                                                                                                                                                                                                                                                                                      | SUBMITTAL NO:<br>N/A                                                                                                                    |
| SCHED. ACTIVITY NO:<br>N/A                                                                                                                                                                                                                                                                                                                   | PAY ITEM NO:<br>N/A                                                                                                                     |
| REQUIREMENT:<br>The ceiling adhesive anchors shall be installed according to the project specifications 723.480-2.08 and manufacturer's requirements so that when the ceiling system is attached to them they do not fail, i.e. pull out.                                                                                                    |                                                                                                                                         |
| DEFICIENCY:<br>At least five ceiling adhesive anchors in the area of ceiling module HOV 079 have failed, that is pull-out has been observed, since the hanging of the ceiling module. See the attached sketch for the locations of the failed anchors and the amount of pull-out that has been observed.                                     |                                                                                                                                         |
| RECOMMENDED ACTION:<br><br>(1) MCC shall contact the adhesive anchor manufacturer to arrange for a site visit and investigation to determine the reason(s) for the failure of the anchors and the proposed corrective actions.<br>(2) MCC shall submit the proposed corrective actions to the Engineer for approval prior to implementation. | RESPONSE DUE DATE: 12/08/99<br><br>RESIDENT ENGINEER: [Signature] DATE: 12-2-99                                                         |
| CORRECTIVE ACTION TAKEN:<br>PLEASE SEE ATTACHED RESPONSE (ACTIONS) BY MCC & LETTER FROM POWERS FASTENERS (SUBMITTAL No 723.480.D14-A)                                                                                                                                                                                                        | FORECAST COMPLETION DATE:<br><br>CONTRACTOR QUALITY CONTROL MANAGER OR APPROPRIATE CONTRACTOR REPRESENTATIVE: [Signature] DATE: 12-6-99 |
| DESIGN CHANGE: YES NO                                                                                                                                                                                                                                                                                                                        | PROJECT ENGINEER: [Signature] DATE:                                                                                                     |
| CORRECTIVE ACTION VERIFIED:                                                                                                                                                                                                                                                                                                                  | ACTUAL COMPLETION DATE:<br><br>RESIDENT ENGINEER: [Signature] DATE:                                                                     |

CA/T 082457



CENTRAL ARTERY/TUNNEL

## Computation Sheet

Subject HOV MOCK-UP CEILING ANCHOR  
BOLT PULL OUT MEASUREMENTS

Page \_\_\_\_\_ of \_\_\_\_\_

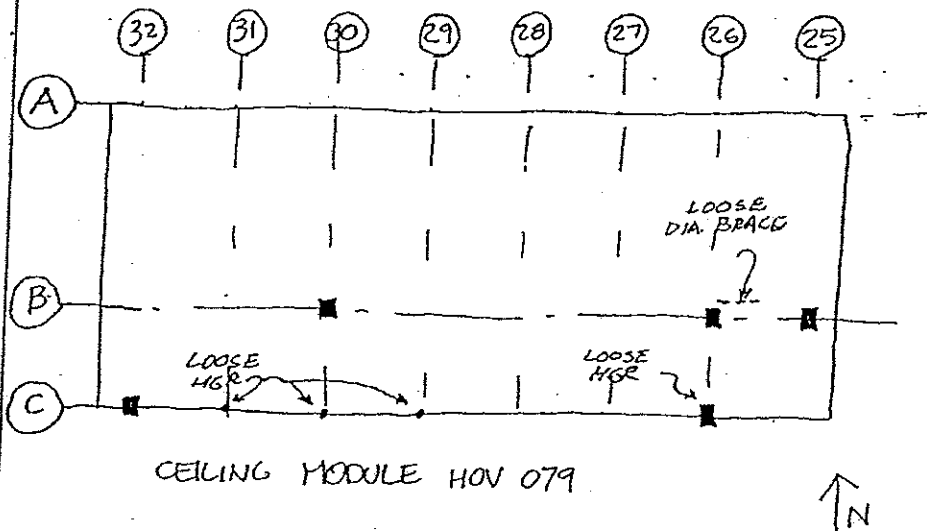
Made by JIM BARRETTDate 10/08/99

Checked by \_\_\_\_\_

Date \_\_\_\_\_

| PLATE<br>LOCATION | CORNER: | GAP BETWEEN TUNNEL ROOF<br>AND CEILING HANGER PLATE |       |       |       |
|-------------------|---------|-----------------------------------------------------|-------|-------|-------|
|                   |         | NE                                                  | NW    | SE    | SW    |
| B-25              |         | 0                                                   | 0     | 7/16" | 7/16" |
| B-26 *            |         | 0                                                   | 3/8"  | 0     | 3/8"  |
| B-30              |         | 1/8"                                                | 3/16" | 0     | 1/16" |
| C-26              |         | 1/4"                                                | 3/16" | 0     | 0     |
| C-32              |         | 0                                                   | 0     | 3/16" | 1/8"  |

\* 4-BOLT (CROSS-BRACING) PL



CA/T 082458

### DR-1 - Adhesive Anchor Failures

In response to the above mentioned Deficiency Report the following corrective actions were / will be taken by MCC upon approval by B/PB.

- 1) On October 21, 1999 a Powers / Rawl Rep. was on site to investigate and determine to reason(s) for the failure of the anchors. Based on information gathered on site, which included a visual inspection of the anchors in question, a determination of failure could not be made.
- 2) MCC submitted a letter / procedure from Powers Fasteners dated Nov. 30, 1999 to B/PB for approval (submittal no 723.480-016-A, also attached). Upon approval MCC will schedule a time to perform and complete this procedure.

CA/T 082459

# **Powers.** **FASTENERS**

Tuesday, November 30, 1999

The Modern Continental Construction Co., Inc.  
6 Necco Court  
Boston MA 02210

Attn: Mr. Neil Joyce  


Powers Fasteners, Inc.  
2 Powers Square  
New Rochelle, NY  
USA 10801

Telephone: 1-(914) 235-8300  
Fax: 1-(914) 576-8483

Web Site: [www.powers.com](http://www.powers.com)  
E-Mail: [williamson@powers.com](mailto:williamson@powers.com)

Re: C09B2 I-90 Seaport Access

Dear Mr. Joyce:

Pursuant to your request, please accept the following suggestions for the repair and replacement of the adhesive anchors where movement was observed on the HOV ceiling mock up.

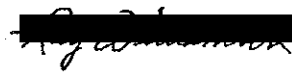
- The adhesive anchor rods in question should be removed from the base material with care, preventing damage to the concrete base material and preserving the integrity of the original anchor location. This can be done with the use of a hydraulic hollow core ram placed directly on the surface to minimize spalling.
- To install replacement adhesive anchors in the same anchor hole, it is recommended to remove any debris from the hole. Upon removal of the threaded rod, the anchor holes should be blown clean with compressed air, brushed clean with a nylon brush, and blown clean again, to ensure that freshly injected adhesive has a solid substrate to bond to.
- The threaded rod removed from the anchor hole should be replaced with a new, clean threaded rod to ensure a good mechanical interlock between the freshly injected adhesive and threads of the anchor rod.
- Adhesive anchor installation procedures, as were previously approved by the Project Engineer and in accordance with Powers Fasteners' published Installation Guidelines should be performed.

CA/T 082460

- Adhesive anchors should be allowed to fully cure. Proof loading per job site specifications should be performed to verify the repair. Upon successful passing of the proof load test, the fixture can be attached to the threaded anchor rod by applying a torque to the nut and washer that does not exceed the published maximum torque range (50 – 90 foot-pounds).

Should you have any questions or require further information, please feel free to contact me at (914) 235-6300. Thank you for your continued interest in Powers Fasteners.

Very truly yours,  
Powers Fasteners, Inc.

  
Ray Williamson  
Engineering Manager

cc: William H. Alderman, P.E. – Powers Fasteners, Inc.  
Jack Armour – Powers Fasteners, New England  
Rich Tamulionis – Newman Renner Colony

CA/T 082461

**Adhesive Anchor Repair Procedure, 12/27/99**



## Adhesive Anchor Repair Procedure

### INDEX STAMP

|                                      |           |
|--------------------------------------|-----------|
| Contract No. 99119-C09B2             | Paragraph |
| Specification Section (Spec-Seq-Rev) |           |
| 723.480 - 016 A                      |           |

### CENTRAL ARTERY / TUNNEL

Permission To Proceed Does Not Constitute Acceptance Or Approval Of Design Details, Calculations, Analysis, Test Methods, Or Materials Developed Or Selected By The Contractor / Supplier And Does Not Relieve The Contractor / Supplier From Full Compliance With Contract Requirements.

|                                     |                                                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/>            | 1. Work May Proceed                                                                                   |
| <input type="checkbox"/>            | 2N. Work May Proceed Subject To The Changes Indicated<br>Resubmittal Not Required                     |
| <input checked="" type="checkbox"/> | 2R. Work May Not Proceed On Indicated Parts<br>Resubmittal Required. Work May Proceed on Other Parts. |
| <input type="checkbox"/>            | 3. Revise and Resubmit. Work May Not Proceed.                                                         |
| <input type="checkbox"/>            | 4. For Information Only.                                                                              |

|          |           |
|----------|-----------|
| REVIEWER |           |
|          | Signature |
|          |           |
|          | Company   |
|          | 12/22/99  |
|          | Date      |

K:\9809\SUBMIT\SUBSTAMP.WK4

CA/T 006797

# **Powers. FASTENERS**

Tuesday, November 30, 1999

The Modern Continental Construction Co., Inc.  
6 Necco Court  
Boston MA 02210

Attn: Mr. Neil Joyca  


**Powers Fasteners, Inc.**  
2 Powers Square  
New Rochelle, NY  
USA 10801

Telephone: 1-(914) 235-8300  
Fax: 1-(914) 576-8483

Web Site: [www.powers.com](http://www.powers.com)  
E-Mail: [williamson@powers.com](mailto:williamson@powers.com)

Re: CO9B2 I-90 Seaport Access

Dear Mr. Joyce:

Pursuant to your request, please accept the following suggestions for the repair and replacement of the adhesive anchors where movement was observed on the HOV ceiling mock up.

- The adhesive anchor rods in question should be removed from the base material with care, preventing damage to the concrete base material and preserving the integrity of the original anchor location. This can be done with the use of a hydraulic hollow core ram placed directly on the surface to minimize spalling.
- To install replacement adhesive anchors in the same anchor hole, it is recommended to remove any debris from the hole. Upon removal of the threaded rod, the anchor holes should be blown clean with compressed air, brushed clean with a nylon brush, and blown clean again, to ensure that freshly injected adhesive has a solid substrate to bond to.
- The threaded rod removed from the anchor hole should be replaced with a new, clean threaded rod to ensure a good mechanical interlock between the freshly injected adhesive and threads of the anchor rod.
- Adhesive anchor installation procedures, as were previously approved by the Project Engineer and in accordance with Powers Fasteners' published Installation Guidelines should be performed.

CA/T 006798

- Adhesive anchors should be allowed to fully cure. Proof loading per job site specifications should be performed to verify the repair. Upon successful passing of the proof load test, the fixture can be attached to the threaded anchor rod by applying a torque to the nut and washer that does not exceed the published maximum torque range (50 – 90 foot-pounds).

Should you have any questions or require further information, please feel free to contact me at (914) 235-6300. Thank you for your continued interest in Powers Fasteners.

Very truly yours,  
Powers Fasteners, Inc.

  
Ray Williamson  
Engineering Manager

cc: William H. Alderman, P.E. – Powers Fasteners, Inc.  
Jack Armour – Powers Fasteners, New England  
Rich Tamulionis – Newman Renner Colony

CA/T 006799



Central Artery/Tunnel

# CONTRACTOR TRANSMITTAL

## OFFICE COPY

TRANSMITTAL DATE:

12/1/99

TRANSMITTAL NO:

27-187

CA/T 006800

CONTRACTOR:

Hobas Construction

ATTENTION:

Robert Sweeney

CONTRACT NO:

99117 - C0732

### DOCUMENT TYPE

- R. REPRODUCIBLE
- P. PRINT
- C. CATALOG CUT
- E. EQUIPMENT MANUAL
- S. SAMPLE
- T. TECHNICAL DATA OR CALCULATIONS

### REVIEW STATUS

- 1. WORK MAY PROCEED.
- 2. WORK MAY PROCEED SUBJECT TO THE CHANGES INDICATED. RESUBMITTAL NOT REQUIRED.
- 3. WORK MAY NOT PROCEED ON INDICATED PARTS. RESUBMITTAL REQUIRED. WORK MAY PROCEED ON OTHER PARTS.
- 4. WORK MAY NOT PROCEED. REVISE AND RESUBMIT.
- 5. FOR INFORMATION ONLY.

DATE RECEIVED BY BIPB RE OFFICE:

12-1-99

DATE DUE BACK TO CONTRACTOR:

12-31-99

DATE SENT BACK TO CONTRACTOR:

12-27-99

RESIDENT ENGINEER:

12-27-99

DATE:

12-27-99

SUBMITTAL NO. (I.E., 120.010 - 011 - A)

| ITEM | SPEC SECT | - SEQ# | - REV | DESCRIPTION                | DOC DATE | DOC TYPE | REVIEW STATUS |
|------|-----------|--------|-------|----------------------------|----------|----------|---------------|
| 1    | 123.470   | 010    | A     | Thickened Asphalt Pavement |          | T        | 27-187        |
| 2    |           |        |       |                            |          |          |               |
| 3    |           |        |       |                            |          |          |               |
| 4    |           |        |       |                            |          |          |               |
| 5    |           |        |       |                            |          |          |               |
| 6    |           |        |       |                            |          |          |               |
| 7    |           |        |       |                            |          |          |               |

COMMENTS TO CONTRACTOR (USE ADDITIONAL SHEETS AS NECESSARY)

REVIEWER:

[Redacted Signature]

DATE:

12/30/99

CR051 (8/99)

WHITE - REVIEWER'S COPY GREEN - RESIDENT ENGINEER'S COPY YELLOW - INFORMATION COPY PINK - PROJECT ENGINEER COPY GOLD - CONTRACTOR'S COPY

PPM -

December 23, 1999  
C09B2 Tunnel Finishes  
DR-001 Adhesive Anchors Failures

The DR-001 proposed fix for the failed anchor bolts, as submitted, is rejected.

22 ( Prior to beginning the repair process the contractor needs to submit documentation on the manufacturer of epoxy and type being used by Powers Fasteners. i.e., If it is a SIKA product what type is it?

The repair/replacement procedure for the failed anchors is as follows:

1. Remove the failed anchors without damaging the hole or adjacent concrete.
2. Use a carbide bit with hammer drill to re-drill/ream the hole to create a new clean surface. There shall be no foreign matter remaining in the hole.
3. Follow manufacturer's procedures to clean and inspect the hole. Begin grouting from the bottom of the hole working towards the surface or opening of the hole. Install a clean, new anchor and tension test the anchor using 125% of the governing Allowable Tension value from the catalog. For a 5/8" dia. Powerfast adhesive anchor the limiting allowable tension value is the adhesive bond strength. The 125% tension test value is 7900 lbs.
4. Install hanger attachment plate and torque anchor bolts to manufacturer's instructions.

Because of evidence concerning installation problems, i.e. dust in the hole and insufficient adhesive in the hole, the capacity of all of the other anchor bolts is in question and will need to be addressed.

There are two options to resolve this problem.

1. Remove all of the anchors and follow the DR procedures to install new bolts, perform tension test on each bolt, reinstall hanger and torque bolts .
2. Retest all of the anchors using 125% of the governing Allowable Tension value, 7900 lbs .

CA/T 006801

**Letter C09B2-0499**



Massachusetts Turnpike Authority  
Central Artery/Tunnel Project

January 7, 2000

Letter No.: C09B2-0499  
File No.: CN-9.09B2.02  
CN-9.09B2.49.01.001

Mr. Robert Coutts  
Project Manager  
Modern Continental Companies, Inc.  
600 Memorial Drive  
Cambridge, MA 02210

Subject: Central Artery (I-93)/Tunnel (I-90) Project  
Contract No. 99119-C09B2  
I-90 Seaport Access Tunnel Finishes  
Deficiency Report (DR) No. 001 - Corrective Action

Dear Mr. Coutts:

Attached for your action is the Project's corrective action for Deficiency Report Number 001 (DR-001), the failure of adhesive anchors in the HOV ceiling. Please incorporate these corrections into the work.

If you have any questions or concerns, please contact me at [REDACTED]

Sincerely,

BECHTEL/PARSONS BRINCKERHOFF

[REDACTED]  
Robert E. Steffy  
Resident Engineer  
Authorized Representative of  
the Project Director

RES/DEH/dh

Attachment: DR-001 Corrective Action (2 pages)

cc: M. Charney  
G. Baxter

CA/T 020256



## DEFICIENCY REPORT

CONTRACT NO: 99119-C09B2 CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES  
DATE OF DEFICIENCY: December 1, 1999 DEFICIENCY REPORT NO: 001  
DATE ISSUED: December 1, 1999 FIELD ENGINEER: P. C. Aikele

|                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| TO:<br>Bob Coutts                                                                                                                                                                                                                                                                                                                              | DISCUSSED WITH:<br>Bill Britto & Jim Martin                                                                                                  |
| SPECIFICATION SECTIONS:<br>723.480 - 2.08                                                                                                                                                                                                                                                                                                      | SUBMITTAL NO:<br>N/A                                                                                                                         |
| SCHED. ACTIVITY NO:<br>N/A                                                                                                                                                                                                                                                                                                                     | PAY ITEM NO:<br>N/A                                                                                                                          |
| <b>REQUIREMENT:</b><br>The ceiling adhesive anchors shall be installed according to the project specifications 723.480-2.08 and manufacturer's requirements so that when the ceiling system is attached to them they do not fail, i.e. pull out                                                                                                |                                                                                                                                              |
| <b>DEFICIENCY:</b><br>At least five ceiling adhesive anchors in the area of ceiling module HOV 079 have failed, that is pull-out has been observed, since the hanging of the ceiling module. See the attached sketch for the locations of the failed anchors and the amount of pull-out that has been observed.                                |                                                                                                                                              |
| <b>RECOMMENDED ACTION:</b><br>(1) MCC shall contact the adhesive anchor manufacturer to arrange for a site visit and investigation to determine the reason(s) for the failure of the anchors and the proposed corrective actions.<br>(2) MCC shall submit the proposed corrective actions to the Engineer for approval prior to implementation | <b>RESPONSE DUE DATE:</b> 12/08/99<br><br>RESIDENT ENGINEER: [Signature] DATE: 12-2-99                                                       |
| <b>CORRECTIVE ACTION TAKEN:</b><br>PLEASE SEE ATTACHED RESPONSE (ACTIONS) BY MCC & LETTER FROM POWERS FASTENERS (SUBMITTAL No 723.460.04-A)                                                                                                                                                                                                    | <b>FORECAST COMPLETION DATE:</b><br>[Signature] DATE: 12-6-99<br>CONTRACTOR QUALITY CONTROL MANAGER OR APPROPRIATE CONTRACTOR REPRESENTATIVE |
| <b>DESIGN CHANGE:</b> YES <input checked="" type="checkbox"/> NO <input type="checkbox"/><br>SEE ATTACHED                                                                                                                                                                                                                                      | <b>PROJECT ENGINEER:</b> [Signature] DATE: 1-6-00<br>[Signature] DATE: 1-7-00                                                                |
| <b>CORRECTIVE ACTION VERIFIED:</b>                                                                                                                                                                                                                                                                                                             | <b>ACTUAL COMPLETION DATE:</b><br><br>RESIDENT ENGINEER: [Signature] DATE: [Signature]                                                       |

CA/T 020257

January 6, 2000

C09B2 Tunnel Finishes

DR-001 Adhesive Anchors Failures

The DR-001 proposed fix for the failed anchor bolts, as submitted, is rejected.

The repair/replacement procedure for the failed anchors is as follows:

1. Remove the failed anchors without damaging the hole or adjacent concrete.
2. Use a carbide bit with hammer drill to re-drill/ream the hole to create a new clean surface. There shall be no foreign matter remaining in the hole. Do not exceed manufacturer's tolerance for hole diameter. If hole diameter exceeds tolerance then change bits, hammer drill a larger diameter hole and install the next larger size anchor and maintain 5 inch hole depth.
3. Follow manufacturer's procedures to clean and inspect the hole. Begin grouting from the bottom of the hole working towards the surface or opening of the hole. Install a clean, new anchor and tension test the anchor using the governing Allowable Tension value from the catalog. For a 5/8" dia. Powerfast adhesive anchor the limiting Allowable Tension value is the adhesive bond strength, 6350 lbs.
4. Install hanger attachment plate and torque anchor bolts to manufacturer's instructions.

Because of evidence concerning installation problems, i.e. dust in the hole and insufficient adhesive in the hole, the capacity of all of the other anchor bolts is in question and will need to be addressed.

There are two options to resolve this problem.

1. Remove all of the anchors and follow the DR procedures to install new bolts. Perform tension test on each bolt using 6350 lbs, reinstall hanger and torque bolts.
2. Retest all of the anchors using the manufacturer's catalog and the governing Allowable Tension for adhesive bond strength - 6350 lbs.

CAIT 020258

**Adhesive Anchor Repair Procedure, 2/16/00**



## Adhesive Anchor Repair Procedure

### INDEX STAMP

|                                      |                  |
|--------------------------------------|------------------|
| Contract No. 99119-C09B2             | Paragraph 3.07 G |
| Specification Section (Spec-Seq-Rev) |                  |
| 723.480 - 016 B                      |                  |

### CENTRAL ARTERY / TUNNEL

Permission To Proceed Does Not Constitute Acceptance Or Approval  
Of Design Details, Calculations, Analysis, Test Methods, Or Materials  
Developed Or Selected By The Contractor / Supplier And Does Not Relieve  
The Contractor / Supplier From Full Compliance With Contract Requirements.

|                                     |                                                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/>            | 1. Work May Proceed                                                                                   |
| <input checked="" type="checkbox"/> | 2N. Work May Proceed Subject To The Changes Indicated<br>Resubmittal Not Required                     |
| <input type="checkbox"/>            | 2R. Work May Not Proceed On Indicated Parts<br>Resubmittal Required. Work May Proceed on Other Parts. |
| <input type="checkbox"/>            | 3. Revise and Resubmit. Work May Not Proceed.                                                         |
| <input type="checkbox"/>            | 4. For Information Only.                                                                              |

REVIEWER

Signature

Company

2/16/00  
Date

K:\9809\SUBMIT\SUBSTAMP.WK4

CA/T 006809

January 28, 2000

Submittal No. 723.480 - 016 B  
Adhesive Anchor Repair Procedure

At this time, the proposed revised anchor testing procedure is applicable to the adhesive anchors installed in the mock-up area in Ramp HOV. ~~The testing parameters described herein are not consistent with those included in Specification Section 723.480, Paragraph 3.07 G. Modern Continental is proceeding in accordance with direction received via Letter No. C09B2-0498, dated January 7, 2000, and reserves our rights to request an equitable adjustment to the Contract, as stipulated in Division I, Section 4.03.~~

AND LETTER No C09B2-0498, JAN 7, 2000

Per the Direction received on the previous submittal, the proposed adhesive anchor repair procedure shall be modified as follows:

1. The failed anchor shall be removed from the concrete without damage to drill hole or adjacent concrete.
2. A hammer drill with a carbide bit shall be utilized to re-drill / ream the hole to create a surface free of debris from the removed anchor. All foreign matter will be removed from the hole.
3. A new adhesive anchor shall be installed per the approved submittal (Submittal 723.480-008 D, Section 3, entitled "Installation Procedure & Quality Control Program - Adhesive Anchoring System").

The new anchor shall be tested in accordance with the directive received in Letter No C09B2-0498, dated January 7, 2000. The revised loading requirement for proof testing shall be at 6,350 lb. tension.

4. The hanger attachment plate will be reinstalled and ~~the~~ bolted to the adhesive anchors. Anchor bolts shall be torqued in accordance with Manufacturer's recommendations.

After completion of the revised testing procedure in the Ramp HOV Mock-Up Area, MCC shall provide test results to the Engineer for review, and await further direction.

*See attached for additional comments*

CA/T 006810

**Submittal Review Comments**  
From Field Engineering Staff  
C09B2

Submittal # 723.480-016 B

Responsible Field Engineer: J. BARRETT

Date review completed and initials: 1/31/00 JB

Field Engineer Comments:

THE PROPOSED REPAIR PROCEDURE SEEMS TO BE WRITTEN  
PERTAINING TO "THE" FAILED ANCHOR AS IF  
THERE IS ONLY ONE. IT SHOULD PERTAIN TO  
"ANY" FAILED ANCHORS.

PER STEP 2 OF PROCEDURE: "ALL FOREIGN  
MATTER WILL BE REMOVED FROM THE HOLE."  
PROCEDURE FOR REMOVING FOREIGN MATTER  
SHOULD INCLUDE ① BLOWING OUT THE HOLE  
WITH COMPRESSED AIR; ② BRUSHING OUT THE  
HOLE (PREFERABLY WITH A STEEL OR BRASS  
WIRE BRUSH—MANUFACTURER RECOMMENDS  
NYLON BRUSH); ③ BLOWING OUT THE HOLE  
AGAIN UNTIL NO DUST REMAINS.

Primary Reviewer BOB RICHARD

Date forwarded to Primary Reviewer \_\_\_\_\_

CAIT 006811

12/31/99 02:56 PM SUBMITTL.RVW